

# ISPyB site report

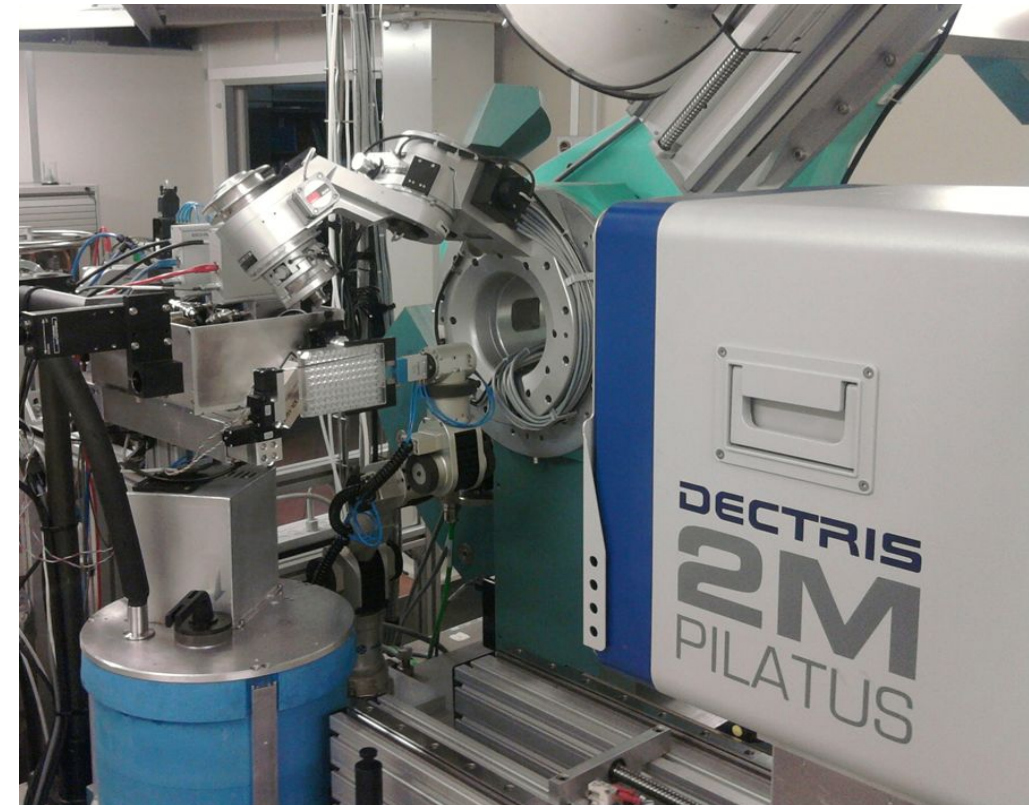
Elettra Sincrotrone Trieste

# Brief beamlines summary (1)

## **XRD1** (X-ray diffraction):

*General purpose XRD - multiple sample supports accepted*

- Huber Kappa Goniometer
- Pilatus 2M
- Arinax BZoom

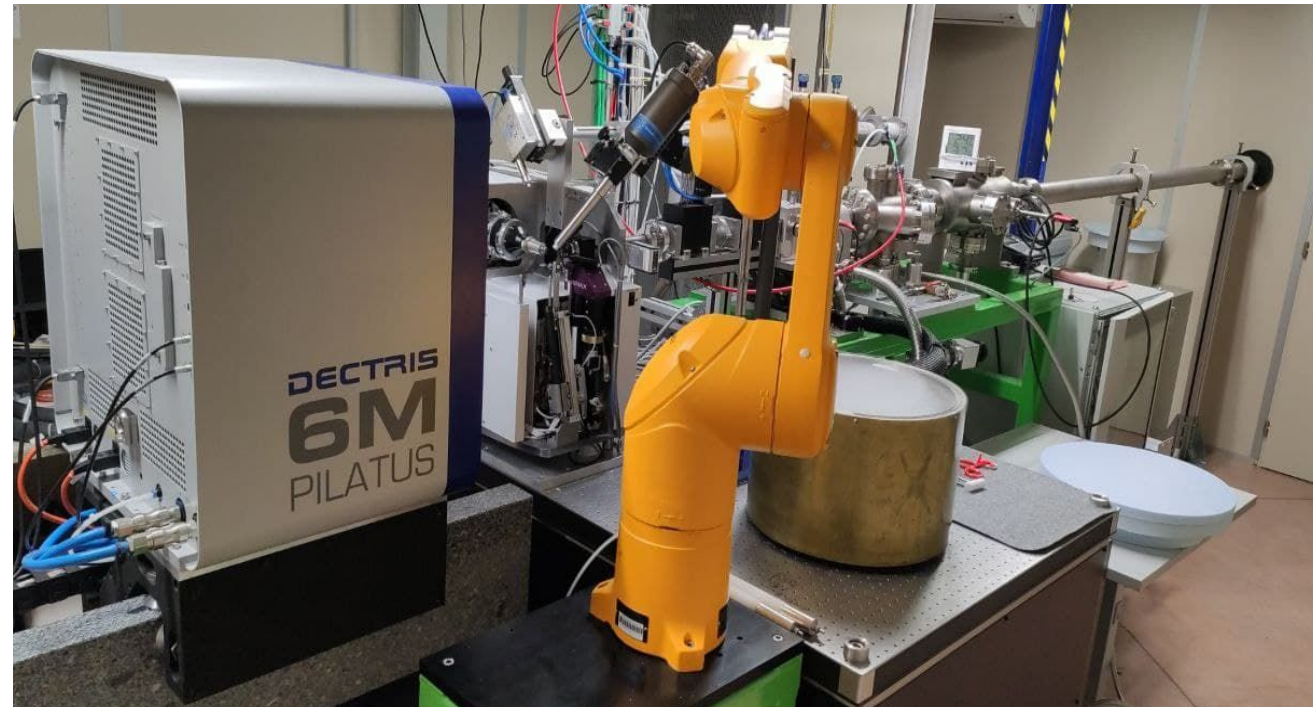


# Brief beamlines summary (2)

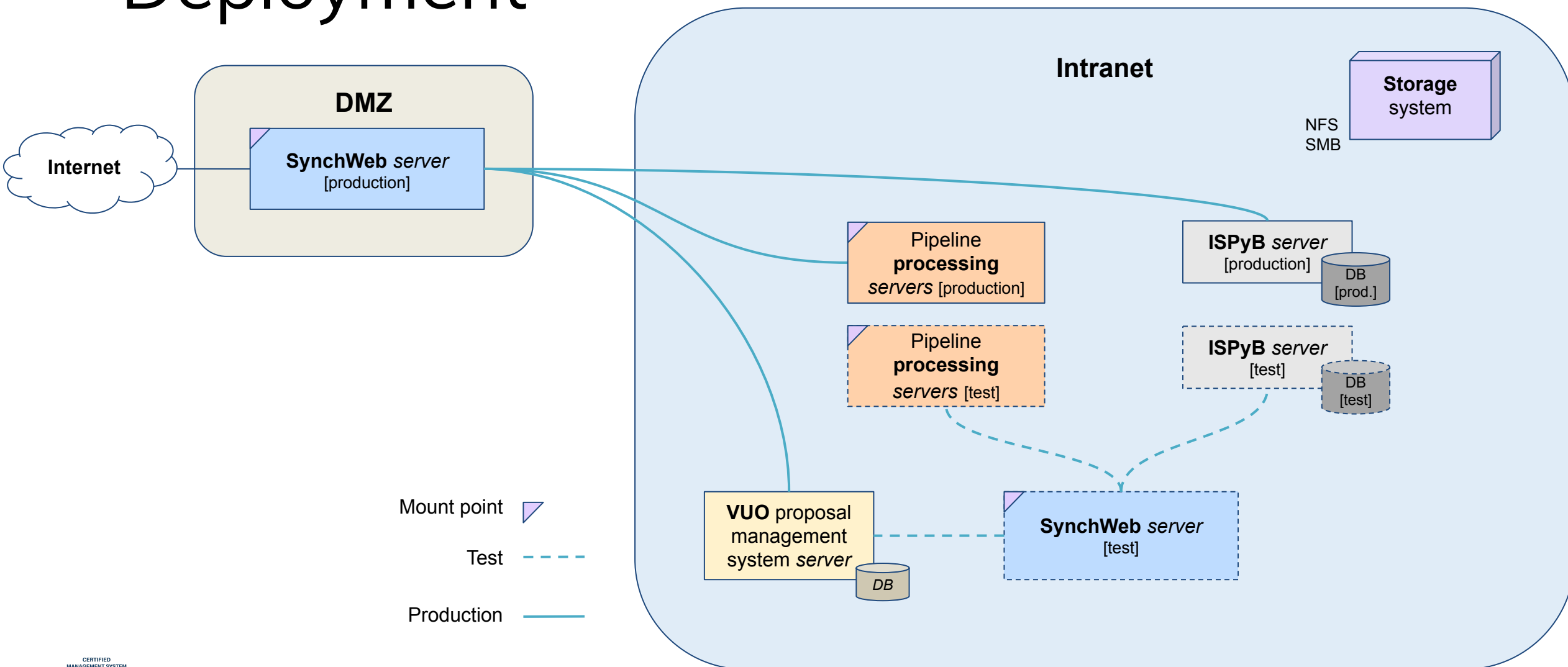
## **XRD2** (X-ray diffraction 2):

*Focussed on MX - SPINE standard samples only*

- Arinax MD2s
- Pilatus 6M
- Arinax BZoom
- Custom sample changer  
(Staubli TX60L on internally  
developed 12 Unipucks dewar  
and in-house developed  
Val3 code)



# Deployment



# ISPyB status (1)

## Backend:

- Deployment at glance:
  - CentOS 7 with Apache in DMZ
  - Schema version **3.0.0**
  - Database version **10.4.12 MariaDB** (DLS version)  
<https://github.com/DiamondLightSource/ispyb-database>

## Frontend:

- Synchweb php 7,
- Minor changes for **interleaved data collection**

# ISPyB status (2)

## Tools:

- **ispyb-datasync**

*Synchronization between VUO and the ISPyB database ingesting all the proposals, users and sessions*

- Python3.6

- **ispyb-api**

*Interface with ISPyB from python*

- Version 9 (compatible with ISPyB v3.0.0)

- **ispyb-database**

- It allows creating and updating the ISPyB schema

# ISPyB and Synchweb

## Questions:

- **SQLAlchemy (ispyb-api)**

We have developed a Python module for MXCuBE 4 (XRD1) that encapsulates all the SQLAlchemy queries needed to interface with ISPyB.

We are now migrating the older module used by MXCuBE 3 (XRD2), which was based on a mix of raw SQL queries and stored procedures

- Are there any performance concerns? (DB size, or complex query) ?
- Would a hybrid approach combining stored procedures and SQLAlchemy queries be a good solution?
- Are there other potential problems we could anticipate?

- **Synchweb :**

- Is there a plan to migrate to PHP 8 ?

# Plans for the next six months

- Allow SynchWeb to relaunch pipelines using RunScriptWeb
- **Dark period**



